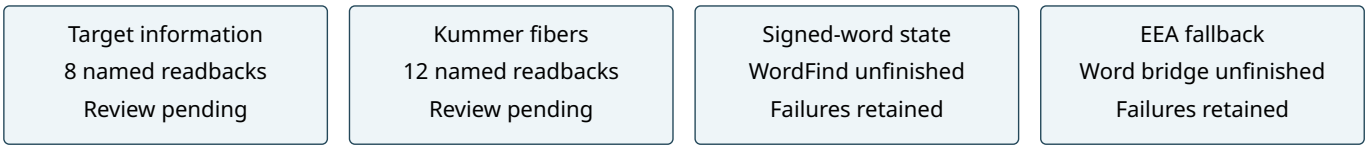
Formal target information and ordinary Kummer fibers

Fresh Lean checks passed eight target-information declarations and twelve Kummer declarations, with exact named axiom inventories and actual false-theorem rejection. Independent source-fidelity review is pending: all three delegated agents terminated with a usage-limit error. The coordinator completed target information and repaired/replayed the saved Kummer source. Signed-word and extended-GCD drafts remain incomplete and their diagnostics are retained.



Target information. For a consistent affine system A x + k b + c = 0 over a field, the target is unique exactly when b lies outside the range of the nuisance linear map A. If b is in that range, the proof constructs a solution for every target value by shifting a known solution. Five main declarations and three controls cover this criterion, a zero coefficient, an inconsistent zero system, and a zero nuisance map. This is the mathematical identifiability contract; native matrix extraction, rank implementation and historical per-edge trajectories are separate interfaces.

Ordinary Kummer fiber. Shared u and b1 squared imply at most two opposite coefficient pairs, including b1=0. The formal source also expands the projection/lift identity: extraction gives b1 squared plus f6 times (a1 squared minus a0) squared. For the quintic source model f6=0, the extra term vanishes. The C1 linear map and normalization connect this identity under injectivity, a correct linear-solve equation and nonzero projective scale. Neighbour orientation then follows under explicit fiber and non-two-torsion-anchor premises. Native solve4 correctness, chart admissibility and geometric transition action remain separate.

Checks. Target information: one fresh theorem module, eight exact readbacks, three native checker rejection controls and false theorem native exit 1. Kummer: four fresh local modules, twelve exact readbacks, three checker rejection controls and false theorem native exit 1. Allowed axioms are propext, Classical.choice and Quot.sound. Source/helper hashes were checked before and after replay. Imported Mathlib and upstream compiled dependencies retain their toolchain/cache trust boundary; no independent binary kernel rechecker ran. This producer packet has no independent fidelity verdict.

Retained failures. Target-information attempt 1 lacked the umbrella tactic import, attempt 2 had an unclosed additive rearrangement, and attempt 3 passed before the final controls were added. Kummer producer source had application-spacing parse errors; the coordinator preserved it and repaired a copy. The first full readback replay used two wrong namespace names and failed; corrected replay 2 passed. No failed readback is counted as verified.

Interrupted lanes. Signed-word potential composition has a saved successful build, while WordFind has unresolved conditional-state and induction proof errors. Its full mutable-word same-root endpoint is not established. The EEA lane has a successful earlier algebra module snapshot and later failed revised/import and WordStep attempts; the current six-register word endpoint and both native fallback calls are not established. These saved intermediate builds are not substituted for the requested endpoint.

Continuation and provenance. Source context is commit 4551ad1ed7. No scientific target, curve enumeration, timing, cloud execution or new cost ratio occurred. The frozen 1,260-target scope and expired campaign authorization are preserved. Diem remains recorded deadline-expired with 50/52 records and no launcher; no duplicate generator was launched. Agent statuses and process inspection showed no remaining compiler job before coordinator continuation. Next: independent review of this producer packet, then complete mutable-word find/edge and EEA/fallback composition under their full contracts.